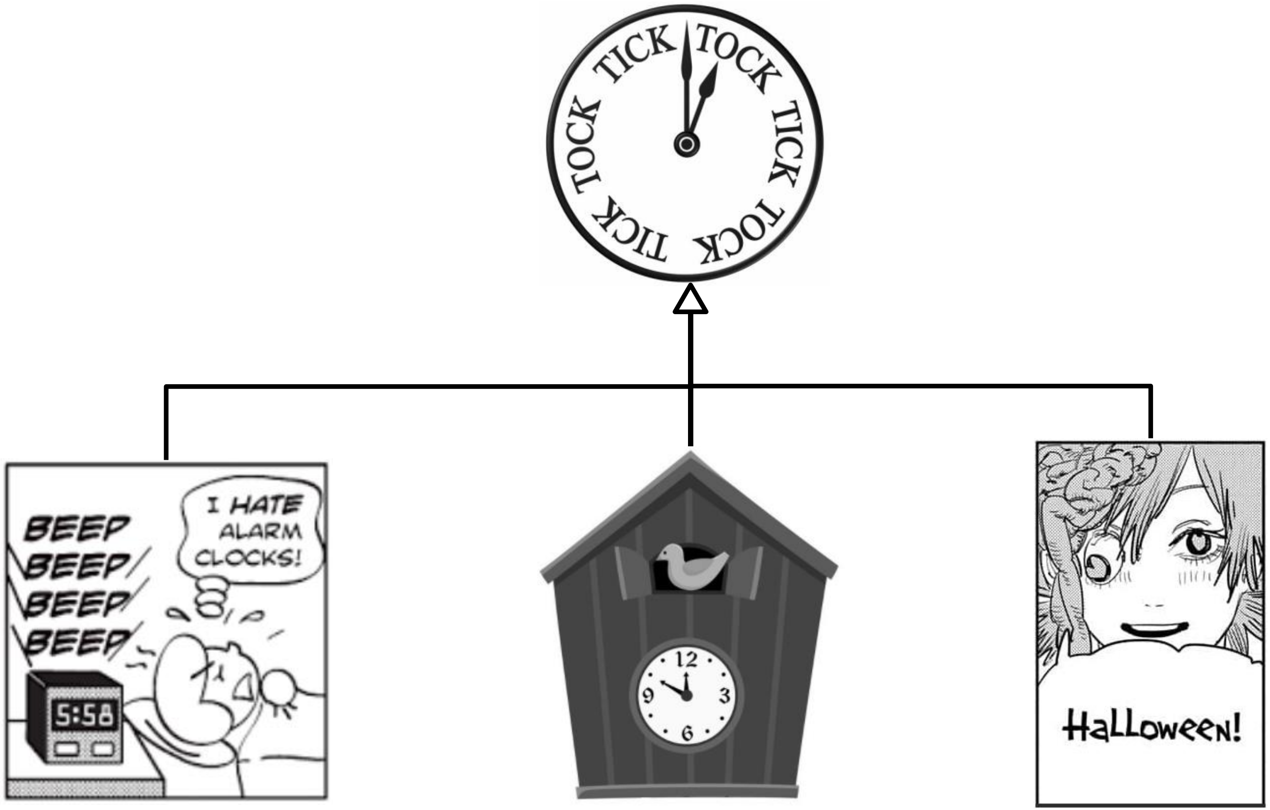

Java Programming

CPT111 – Week 8

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CPT111 Java Programming

Week 8 Exercise and Coursework 1

More Objects and Inheritance

Coding and Submission

- Coding in your NetBeans
  - Start with the skeleton code given in the course LMO

- Submitting into Learning Mall Quiz
  - Do not submit the whole class
  - Only submit the constructor or the method
    - read carefully the instructions
  - You can submit your own private helper method
    - but *do not* add another public methods

Clock + Getters

- In the exercises Week 8
  - we will subclass the class Clock from Week 6
  - but with **<u>you must add these getters into it</u>**

```java
// Getters
public int getHours() {
    return hours;
}
public int getMinutes() {
    return minutes;
}
```

  - complete Clock.java with your Week 6 code ***first***!

Alarm Clock

- AlarmClock is a subclass of Clock
  - you can still use the inherited methods such as toString()

- In addition, you can now also set up an alarm to alert you in a specified time
  - using the default or a chosen message
  - by overriding Clock's method tick()

Exercise #8.1 AlarmClock Constructor 1

- Complete the first constructor of the class AlarmClock
- It takes four arguments: h, m, alarmHours, alarmMinutes
  - creates a new AlarmClock object whose initial time is h hours and m minutes
  - sounds an alarm at alarmHours hours and alarmMinutes minutes, with the default sound "Beep beep beep beep!"

- Test cases:

```
AlarmClock ac1 = new AlarmClock(5, 58, 6, 0);
ac1.tick();
ac1.tick();                              → Beep beep beep beep!
System.out.println(ac1);                 → 06:00
```

Exercise #8.2 AlarmClock Constructor 2

- Complete the second constructor of the class AlarmClock
- Overloading the first constructor, it now takes five arguments: h, m, alarmHours, alarmMinutes, and alarmSound
  - creates a new AlarmClock object whose initial time is h hours and m minutes
  - sounds an alarm at alarmHours hours and alarmMinutes minutes, and sets the sound to alarmSound
- Test cases:

```
AlarmClock ac2 = new AlarmClock(14, 29, 14, 30, "Wake Up! The Hero! Kamen Rider!");
ac2.tick();                                → Wake Up! The Hero! Kamen Rider!
```

Exercise #8.3 AlarmClock Tick

- Complete the method tick of the class AlarmClock
- It overrides the method tick of Clock and adds 1 minute to the time on this alarm clock
  - In addition, it sounds (prints) the alarm at the specified time.

- Test cases:

```
AlarmClock ac1 = new AlarmClock(5, 58, 6, 0);
ac1.tick();
ac1.tick();                          → Beep beep beep beep!
System.out.println(ac1);             → 06:00
AlarmClock ac2 = new AlarmClock(14, 29, 14, 30, "Wake Up! The Hero! Kamen Rider!");
ac2.tick();                          → Wake Up! The Hero! Kamen Rider!
```

Cuckoo Clock

- CuckooClock is a subclass of Clock
  - you can still use the inherited methods such as toString()

- In addition, the CuckooClock can print "Cuckoo!" at the start of every hour
  - as many as the current hours, no matter it is morning or night

Exercise #8.4 CuckooClock Constructor

- Complete the constructor of the class CuckooClock
- It takes two arguments: h and m
  - and creates a new CuckooClock object whose initial time is h hours and m minutes

- Test cases:

```
CuckooClock cc1 = new CuckooClock(0, 58);
cc1.tick();
cc1.tick();                                   → Cuckoo!
System.out.println(cc1);                      → 01:00
CuckooClock cc2 = new CuckooClock(13, 59);
cc2.tick();                                   → Cuckoo!↵Cuckoo!
```

Exercise #8.5 CuckooClock Tick

- Complete the method tick of the class CuckooClock
- It overrides the method tick of Clock and adds 1 minute to the time on this Cuckoo clock
- In addition, it prints "Cuckoo!" at the start of every hour
  - It prints one time for each hour
  - Whether it is morning or night does **not** change the number of times it prints

- Test cases:

```
CuckooClock cc1 = new CuckooClock(0, 58);
cc1.tick();
cc1.tick();                                   → Cuckoo!
System.out.println(cc1);                      → 01:00
CuckooClock cc2 = new CuckooClock(13, 59);
cc2.tick();                                   → Cuckoo!↲Cuckoo!
```

Halloween Clock

- HalloweenClock is a subclass of Clock
  - you can still use the inherited methods such as toString()

- In addition, whenever ***any*** current instances of HalloweenClock have ticked three times
  - the latest ticking object will also print "Halloween!"

Exercise #8.6 HalloweenClock Constructor

- Complete the constructor of the class HalloweenClock
- It takes two arguments: h and m
  - and creates a new HalloweenClock object whose initial time is h hours and m minutes
- Test cases:

```
HalloweenClock hc1 = new HalloweenClock(1, 0);
HalloweenClock hc2 = new HalloweenClock(2, 0);
hc1.tick();
hc2.tick();
hc2.tick();                    → Halloween!
HalloweenClock hc3 = new HalloweenClock(3, 30);
hc1.tick();
hc2.tick();
hc3.tick();                    → Halloween!
System.out.println(hc3);       → 03:31
```

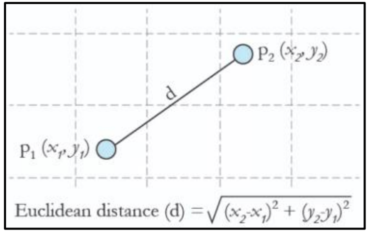
Exercise #8.7 HalloweenClock Tick

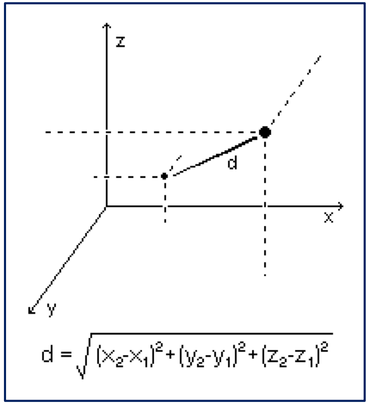
- Complete the method tick of the class HalloweenClock
- It overrides the method tick of Clock, adds 1 minute to the time on this Halloween clock
  - and if ***any*** Halloween clocks have ticked three times, prints "Halloween!"
- Test cases:

```
HalloweenClock hc1 = new HalloweenClock(1, 0);
HalloweenClock hc2 = new HalloweenClock(2, 0);
hc1.tick();
hc2.tick();
hc2.tick();                    → Halloween!
HalloweenClock hc3 = new HalloweenClock(3, 30);
hc1.tick();
hc2.tick();
hc3.tick();                    → Halloween!
System.out.println(hc3);       → 03:31
```

CW1 Week #8

- Complete the class **ThreeDimPoint** that represents a point in the 3-dimensional space
  - It extends the class **EuclideanPoint** in Week 6
  - A skeleton file ThreeDimPoint.java is given





CW1 #8.1 EuclideanPoint distanceTo

- Complete the method **distanceTo** of the class **ThreeDimPoint**, a subclass of the class **EuclideanPoint**
- Some code is already given
- Given an instance of ThreeDimPoint, it first cast the other point to ThreeDimPoint variable, then it computes and returns the distance between this point and the other point in the parameter
- Otherwise, it calls the overridden method of the superclass
- Test cases:

```
ThreeDimPoint point1 = new ThreeDimPoint(1.0, 2.0, 3.0);
ThreeDimPoint point2 = new ThreeDimPoint(3.0, 5.0, 9.0);
System.out.println(point1.distanceTo(point2));          → 7.0
EuclideanPoint point3 = new EuclideanPoint(8.0, 26.0);
System.out.println(point1.distanceTo(point3));          → 25.0
```

Thank you for your attention!

- This is the end of Week 8 Exercise and Coursework 1 Task Sheet